PART 1 / STATIC LONGITUDINAL STABILITY

From aircraft trim to static stability

Week 4 found the elevator and tab settings for balanced flight. The next question is whether a small incidence disturbance produces a restoring pitching moment.

$$C_M = 0 \quad \text{at trim,} \qquad \Delta\alpha > 0 \quad \Longrightarrow \quad \Delta C_M < 0 \quad \text{for a restoring response.}$$

Trim	Forces and moments balance at the chosen flight condition.
Static stability	A small disturbance produces a moment opposing that disturbance.

The sequence in this part is:

1. Define the condition for longitudinal static stability.
2. Derive the stick-fixed and stick-free CG margins.
3. Calculate both margins using the stated aircraft data.
4. Relate the margins to elevator and hinge-torque trim gradients.

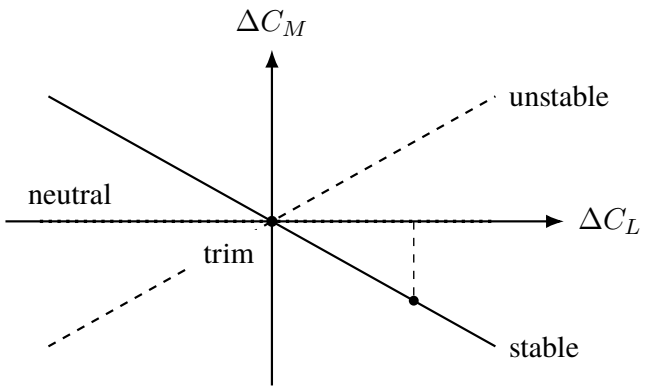
Part 2 connects these gradients to stick movement and pilot force.

1 / THE STABILITY TEST

The condition for longitudinal static stability

Start from trim. Increase wing incidence α slightly, without retrimming. If the air-relative flight-path direction is initially unchanged, a small nose-up rotation gives this increase. From Week 2, i_w is the fixed wing no-lift-line setting relative to the fuselage datum:

$$\alpha = \theta + i_w - \gamma, \qquad \Delta\gamma \simeq 0 \implies \Delta\alpha \simeq \Delta\theta.$$



more lift → nose-down increment on the stable line

Increment coordinates: the origin represents the initial trim point. Positive C_M is nose-up. Controls must obey the stated fixed or free constraint.

$$\Delta C_M \simeq \frac{dC_M}{d\alpha}\Delta\alpha, \qquad \Delta C_L \simeq \frac{dC_L}{d\alpha}\Delta\alpha.$$

For a positive lift slope, the two equivalent tests are

$$\frac{dC_M}{d\alpha} < 0 \qquad \Longleftrightarrow \qquad \frac{dC_M}{dC_L} < 0.$$

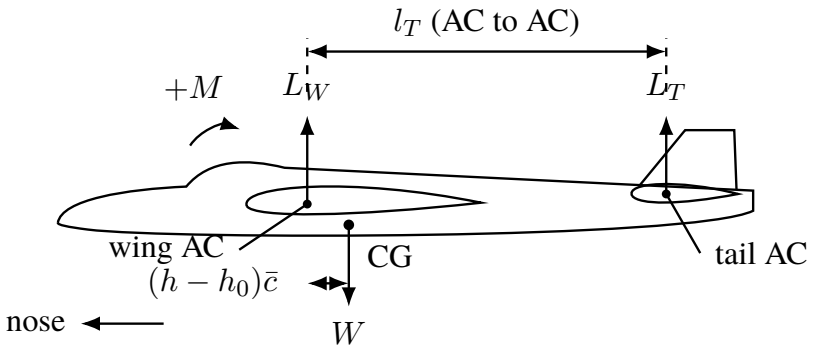
At fixed speed and density, $dM/dL = \bar{c}\, dC_M/dC_L$ has the same sign. A negative incidence change then produces a nose-up restoring moment.

This is a **static tendency**. Whether the later motion decays or oscillates requires inertia, damping and the dynamic equations. Zero slope is neutral in this first-order test.

1 / CG POSITION AND PITCHING MOMENT

Start with the wing and tail moment balance

h is the aircraft CG position; h_0 is the wing AC position. Both are measured aft from the mean aerodynamic chord's leading edge and divided by $\bar{c}$.



$$\bar{V}_T = \frac{S_T l_T}{S_W \bar{c}}, \qquad C_M = C_{M0} + (h - h_0)C_L - \bar{V}_T C_L^T.$$

Week 3 retained both lift contributions. The original Week 5 model uses

$$C_L = C_L^W + \frac{S_T}{S_W} C_L^T \simeq C_L^W = a\alpha, \qquad a = a_w.$$

Tail lift is retained in the moment balance. From Weeks 2–4,

$$\alpha_T = \alpha + \eta_T - \epsilon = \frac{C_L}{a}\left(1 - \frac{d\epsilon}{d\alpha}\right) + \eta_T.$$

$$C_L^T = a_1\alpha_T + a_2\eta + a_3\beta,$$
$$C_H = b_1\alpha_T + b_2\eta + b_3\beta.$$

α uses the wing no-lift line; α_T (also written α^T) uses the tail no-lift line. η_T is tail setting; η is elevator deflection; $\beta = \beta_{\text{tab}}$ is tab deflection relative to the elevator. Positive control angles are trailing edge down. The a_i, b_i are per-radian derivatives.

2 / STICK-FIXED STABILITY

Hold the elevator and find the restoring slope

Constraint: elevator η and tab β are fixed during the incidence disturbance. Tail setting and aerodynamic derivatives also stay fixed. Tail incidence and tail lift can still change.

$$C_M = C_{M0} + (h - h_0)C_L \\ - \bar{V}_T \left[\frac{a_1 C_L}{a} \left(1 - \frac{d\epsilon}{d\alpha} \right) + a_1 \eta_T + a_2 \eta + a_3 \beta \right].$$

Constant terms have zero derivative; the derivative of each linear C_L term is its multiplying coefficient:

$$\left. \frac{dC_M}{dC_L} \right|_{\eta,\beta} = (h - h_0) - \bar{V}_T \frac{a_1}{a} (1 - \frac{d\epsilon}{d\alpha}).$$

Define the **stick-fixed CG margin**, also called the static margin:

$$H_n \equiv - \left. \frac{dC_M}{dC_L} \right|_{\eta,\beta} = (h_0 - h) + \bar{V}_T \frac{a_1}{a} (1 - \frac{d\epsilon}{d\alpha}).$$

The corresponding **neutral point** is the CG location giving zero slope:

$$h_n = h_0 + \bar{V}_T \frac{a_1}{a} (1 - \frac{d\epsilon}{d\alpha}), \qquad H_n = h_n - h.$$

CG position	Margin	Static response
$h < h_n$	$H_n > 0$	restoring
$h = h_n$	$H_n = 0$	neutral
$h > h_n$	$H_n < 0$	destabilising

Increasing tail volume strengthens the restoring contribution when $a_1 > 0$ and $\frac{d\epsilon}{d\alpha} < 1$. Moving the CG aft reduces the margin.

2 / STICK-FREE CG MARGIN

Let the elevator float and differentiate

The tab stays fixed. The elevator is free to settle at zero aerodynamic hinge torque:

$$0 = b_1\alpha_T + b_2\eta + b_3\beta \quad \Longrightarrow \quad \eta = -\frac{b_1}{b_2}\alpha_T - \frac{b_3}{b_2}\beta.$$

Substitute into the tail lift law and collect terms:

$$\begin{aligned} C_L^T &= a_1\alpha_T + a_2\left(-\frac{b_1}{b_2}\alpha_T - \frac{b_3}{b_2}\beta\right) + a_3\beta \\ &= \bar{a}_1\alpha_T + \bar{a}_3\beta, \\ \bar{a}_1 &= a_1 - \frac{a_2 b_1}{b_2}, \qquad \bar{a}_3 = a_3 - \frac{a_2 b_3}{b_2}. \end{aligned}$$

$$\begin{aligned} C_{M,\text{free}} = C_{M0} &+ (h - h_0)C_L \\ &- \bar{V}_T\left[\frac{\bar{a}_1 C_L}{a}\left(1 - \frac{d\epsilon}{d\alpha}\right) + \bar{a}_1\eta_T + \bar{a}_3\beta\right]. \end{aligned}$$

At fixed tab, the terms independent of C_L disappear on differentiation:

$$\left.\frac{dC_M}{dC_L}\right|_{C_H=0,\beta} = (h - h_0) - \bar{V}_T\frac{\bar{a}_1}{a}\left(1 - \frac{d\epsilon}{d\alpha}\right).$$

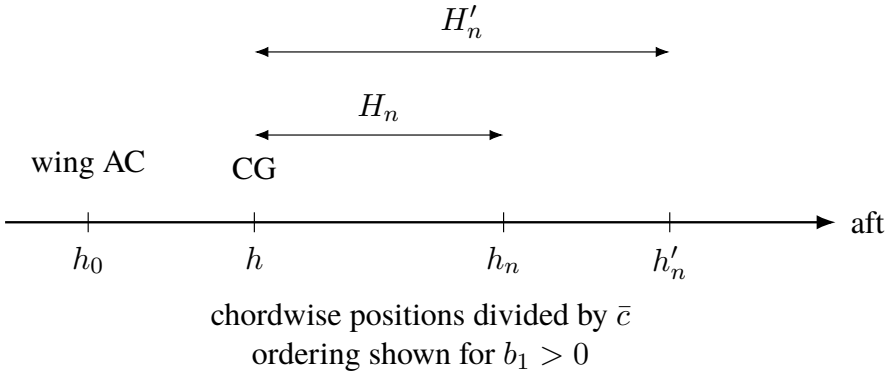
$$H_n' = -\frac{dC_M}{dC_L} = (h_0 - h) + \bar{V}_T\frac{\bar{a}_1}{a}\left(1 - \frac{d\epsilon}{d\alpha}\right).$$

The prime denotes stick-free. $H_n' > 0$ is restoring; $H_n' = 0$ is neutral. The bar includes elevator movement; it does not mean $\eta = 0$.

2 / COMPARING THE TWO MARGINS

The hinge derivatives decide which margin is larger

Both margins measure the CG distance ahead of a neutral point. The neutral point represents the *incremental* lift response under its specified control constraint.



Subtract the stick-fixed margin from the stick-free margin:

$$H'_n - H_n = \frac{\bar{V}_T}{a}\left(1 - \frac{d\epsilon}{d\alpha}\right)(\bar{a}_1 - a_1) = -\frac{\bar{V}_T}{a}\left(1 - \frac{d\epsilon}{d\alpha}\right)\frac{a_2 b_1}{b_2}.$$

For an aft tail, take $\bar{V}_T, a, a_2 > 0$, $\frac{d\epsilon}{d\alpha} < 1$ and $b_2 < 0$:

Hinge response	Effective tail slope	Margin ordering
$b_1 > 0$	$\bar{a}_1 > a_1$	$H'_n > H_n$
$b_1 = 0$	$\bar{a}_1 = a_1$	$H'_n = H_n$
$b_1 < 0$	$\bar{a}_1 < a_1$	$H'_n < H_n$

For $b_1 > 0$, increased tail incidence floats the elevator down and adds to the tail lift increment. For $b_1 < 0$, it floats up and reduces that increment.

The ordering follows from the derivatives. Releasing an elevator does not universally increase or decrease static stability.

2 / WORKED EXAMPLE

Calculate both margins for the same aircraft

Question. A trimmed aircraft has the following data. Is its longitudinal response statically stable with the elevator held and with it free?

$$\bar{V}_T = 0.8, \quad h - h_0 = 0.1, \quad \frac{d\epsilon}{d\alpha} = 0.4, \quad a = 5.7\,1/\mathrm{rad},$$

$$a_1 = 5.0, \quad a_2 = 4.0, \quad a_3 = 0.5, \qquad b_1 = 0.05, \quad b_2 = -0.10, \quad b_3 = -0.05.$$

All a_i and b_i have units $1/\mathrm{rad}$. Use the compact lift approximation introduced above.

$$\begin{aligned} H_n &= (h_0 - h) + \bar{V}_T \frac{a_1}{a}(1 - \frac{d\epsilon}{d\alpha}) \\ &= -0.1 + \frac{0.8 \cdot 5.0}{5.7} \cdot (1 - 0.4) = \boxed{0.321}, \\ \bar{a}_1 &= a_1 - \frac{a_2 b_1}{b_2} = 5.0 - \frac{4.0 \cdot 0.05}{-0.10} = \boxed{7.0\,1/\mathrm{rad}}, \\ H_n' &= (h_0 - h) + \bar{V}_T \frac{\bar{a}_1}{a}(1 - \frac{d\epsilon}{d\alpha}) \\ &= -0.1 + \frac{0.8 \cdot 7.0}{5.7} \cdot (1 - 0.4) = \boxed{0.489}. \end{aligned}$$

Both margins are positive. This coefficient set gives a larger free margin because $b_1 > 0$.

For a small increase $\Delta C_L = 0.10$: $\Delta C_{M,\mathrm{fixed}} \simeq -0.032$ and $\Delta C_{M,\mathrm{free}} \simeq -0.049$. Both moments are nose-down.

a_3 and b_3 affect the trim settings. They do not enter these slopes because the tab is fixed during the disturbance.

3 / FROM STABILITY TO TRIM

The fixed margin sets the elevator trim gradient

Now compare a **sequence of trimmed conditions**: $C_M = 0$ at every point. Hold the tab and configuration fixed, but adjust the elevator as the required C_L changes.

$$\begin{aligned} 0 = {} & C_{M0} + (h - h_0)C_L \\ & - \bar{V}_T \left[\frac{a_1 C_L}{a} \left(1 - \frac{d\epsilon}{d\alpha} \right) + a_1 \eta_T + a_2 \eta_{\text{trim}} + a_3 \beta \right]. \end{aligned}$$

Collect the elevator term and the terms multiplying C_L:

$$\begin{aligned} \bar{V}_T a_2 \eta_{\text{trim}} = {} & C_{M0} - \bar{V}_T (a_1 \eta_T + a_3 \beta) \\ & + \left[(h - h_0) - \bar{V}_T \frac{a_1}{a} (1 - \frac{d\epsilon}{d\alpha}) \right] C_L. \end{aligned}$$

The bracket is $-H_n$. Therefore

$$\eta_{\text{trim}} = \underbrace{\frac{C_{M0}}{\bar{V}_T a_2} - \frac{a_1 \eta_T}{a_2} - \frac{a_3 \beta}{a_2}}_{\eta_0} - \frac{H_n}{\bar{V}_T a_2} C_L.$$

$$\frac{d\eta_{\text{trim}}}{dC_L} = -\frac{H_n}{\bar{V}_T a_2}.$$

For $H_n > 0$, $\bar{V}_T > 0$ and $a_2 > 0$, a larger required lift coefficient needs a more negative elevator angle: **more up elevator**.

This gradient joins equilibria. The stability slope on the stick-fixed derivation described a disturbance with the elevator held; it did not set $C_M = 0$ again after each change.

3 / HINGE-TORQUE TRIM GRADIENT

Substitute the trimmed elevator into the hinge law

Along this curve the aircraft is trimmed, while the elevator is held at the required angle. Its hinge torque need not be zero. Insert the complete elevator trim equation from page 8 into the hinge law:

$$C_{H,\mathrm{trim}} = b_1\left[\frac{C_L}{a}\left(1-\frac{d\epsilon}{d\alpha}\right)+\eta_T\right]+b_3\beta + b_2\left[\frac{C_{M0}}{\bar{V}_T a_2}-\frac{a_1\eta_T}{a_2}-\frac{a_3\beta}{a_2}-\frac{C_L H_n}{\bar{V}_T a_2}\right].$$

Collect the setting and tab terms:

$$\bar{b}_1 = b_1-\frac{b_2 a_1}{a_2}, \qquad \bar{b}_3 = b_3-\frac{b_2 a_3}{a_2}.$$

$$C_{H,\mathrm{trim}} = \frac{b_2 C_{M0}}{\bar{V}_T a_2}+\bar{b}_1\eta_T+\bar{b}_3\beta + \left[\frac{b_1}{a}\left(1-\frac{d\epsilon}{d\alpha}\right)-\frac{b_2 H_n}{\bar{V}_T a_2}\right]C_L.$$

Factor $b_2/(\bar{V}_T a_2)$ from the coefficient of C_L:

$$\frac{b_1}{a}\left(1-\frac{d\epsilon}{d\alpha}\right)-\frac{b_2 H_n}{\bar{V}_T a_2} = \frac{b_2}{\bar{V}_T a_2}\left[(h-h_0)-\bar{V}_T\frac{\bar{a}_1}{a}\left(1-\frac{d\epsilon}{d\alpha}\right)\right] = -\frac{b_2 H'_n}{\bar{V}_T a_2}.$$

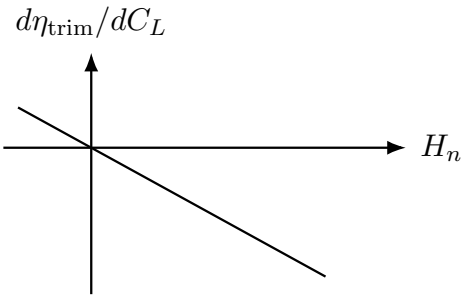
$$C_{H0} = \frac{b_2 C_{M0}}{\bar{V}_T a_2}+\bar{b}_1\eta_T+\bar{b}_3\beta.$$

$$C_{H,\mathrm{trim}} = C_{H0}-\frac{b_2 C_L H'_n}{\bar{V}_T a_2}, \qquad \frac{dC_{H,\mathrm{trim}}}{dC_L} = -\frac{b_2 H'_n}{\bar{V}_T a_2}.$$

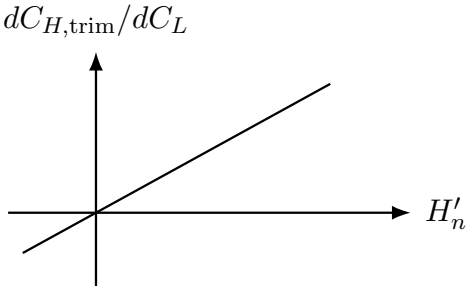
The free margin appears in the holding-torque gradient. This does not impose $C_H = 0$ along the whole trim curve.

3 / READING THE GRADIENTS

Two trim gradients reveal two different margins



positive fixed margin
negative elevator slope



positive free margin
positive hinge-coefficient slope

Each point represents a different stability margin, with the other factors held fixed. These are plots of a gradient against margin, not elevator angle or torque against time.

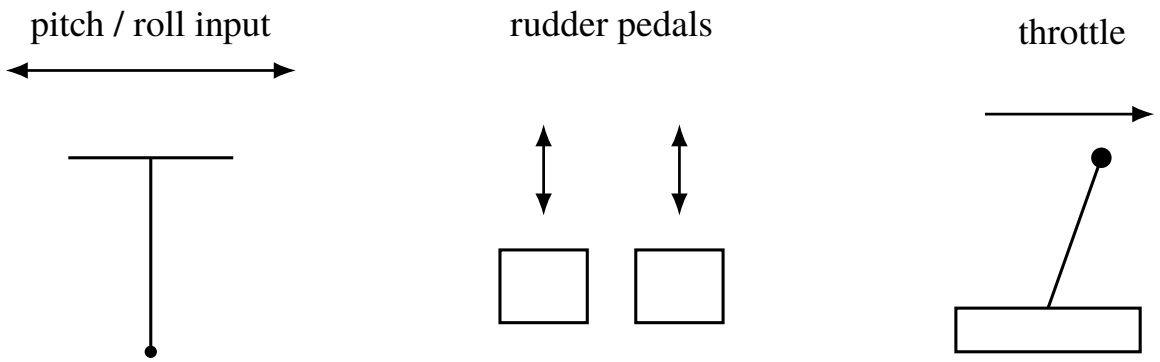
Elevator trim gradient	Hinge-coefficient trim gradient
$\dfrac{d\eta_{\text{trim}}}{dC_L} = -\dfrac{H_n}{\bar{V}_T a_2}$	$\dfrac{dC_{H,\text{trim}}}{dC_L} = -\dfrac{b_2 H'_n}{\bar{V}_T a_2}$
Zero when $H_n = 0$.	Zero when $H'_n = 0$.
Units: radians per unit C_L.	Dimensionless coefficient per unit C_L.

For an aft tail with $a_2 > 0$ and $b_2 < 0$, moving the CG aft reduces both margins. The elevator gradient becomes less negative; the hinge-coefficient gradient becomes less positive.

A zero crossing identifies the neutral condition for that particular constraint. An aircraft can lose one positive margin while retaining the other.

4 / AIRCRAFT CONTROL

Stick, pedals and throttle have distinct roles



Pilot input	Conventional surface action	Main initial effect
Push forwards	Elevator trailing edge down	Nose-down pitching moment
Pull backwards	Elevator trailing edge up	Nose-up pitching moment
Stick / wheel right	Right aileron up; left aileron down	Right wing down
Right pedal	Rudder trailing edge right	Nose yaws right
Throttle forwards	Increase engine thrust or power command	Change energy input

For the conventional aft tail, lowering the elevator increases tail lift. The upward force behind the CG produces a nose-down moment.

Part 2 will relate the stick positions needed at *different steady speeds* to the trim gradients derived here.

Throttle does not control speed independently of everything else. Speed, climb rate, drag and pitch trim interact. Control systems can also modify the direct response shown here.

Conclusions

Static longitudinal stability requires a restoring pitching-moment increment after a small incidence disturbance. For a positive lift slope:

$$\frac{dC_M}{dC_L} < 0.$$

$$H_n = (h_0 - h) + \bar{V}_T \frac{a_1}{a}\left(1 - \frac{d\epsilon}{d\alpha}\right),$$
$$H'_n = (h_0 - h) + \bar{V}_T \frac{\bar{a}_1}{a}\left(1 - \frac{d\epsilon}{d\alpha}\right),$$
$$\bar{a}_1 = a_1 - \frac{a_2 b_1}{b_2}.$$

Positive margins are restoring. Zero margin is neutral. Moving the CG aft reduces both margins at fixed geometry.

$$\frac{d\eta_{\text{trim}}}{dC_L} = -\frac{H_n}{\bar{V}_T a_2}, \qquad \frac{dC_{H,\text{trim}}}{dC_L} = -\frac{b_2 H'_n}{\bar{V}_T a_2}.$$

For an aft tail with $a_2 > 0$ and $b_2 < 0$, positive margins give a negative elevator trim gradient and a positive hinge-coefficient trim gradient.

> Part 2 uses these two gradients to find how the stick position and pilot force change with steady flight speed.

Tutorial 3, questions 4–6, apply the margin and tab-trim relationships.